Lower α Reveals Tail Uncertainty

Formula relationship → weighting mechanism → empirical error detection

1 DEFINE

Connect Rényi to Shannon

Rényi entropy family

$$H_\alpha(p) = \frac{1}{1-\alpha} \log \sum_v p(v)^\alpha$$

α → 1

$$\lim_{\alpha \to 1} H_\alpha(p) = H(p) = -\sum_v p(v) \log p(v)$$

α → 1 recovers Shannon entropy.

One parameter controls which probabilities matter.

2 REWEIGHT

See what α changes

Same probabilities, different emphasis

Effective weight ∝ p(v)^α

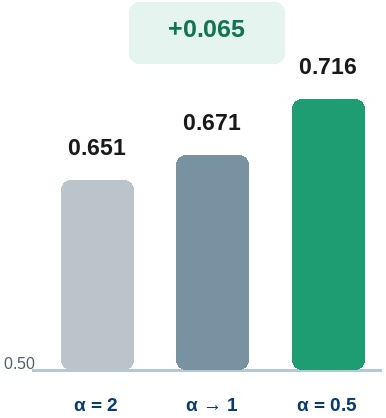


Lower α → stronger tail influence

3 DETECT

Measure the empirical effect

2Wiki error-detection AUROC ↑



Tail-sensitive α = 0.5 performs best.

As α decreases, tail influence and error-detection AUROC both increase.